Diversity Reduction in Activation Steering — Walk-Through

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Reading Guide

This document develops a theoretical account of why activation steering collapses the diversity of a transformer's outputs, and tests the account empirically on two open-weight models. It is written so a reader with no prior exposure to the project can follow it linearly. The structure is:

1. **First-order theory** (Section 1). Taylor-expand the RMSNorm projection around the steered mean. Result: an explicit formula for the post-normalization covariance, with two terms — a rank-reducing projection orthogonal to the steered mean direction, and a variance-shrinking factor of $1/\|\mu + s\|^2$.

2. **Global geometric bounds** (Section 1.6). Use only an elementary chordal-contraction lemma to derive distribution-free upper bounds on spherical variance, expected pairwise distance, and spherical diameter. These bounds are exact, not approximations.

3. **Second-order corrections** (Section 1.7). Refine the Taylor expansion to second order. Two new effects appear: sphere curvature pulls the mean resultant length below 1, and a radial-perpendicular coupling encodes the magnifying-glass effect near the origin. The refined formula is what predicts the parabolic shape we observe in the empirical $V(s)$ curves.

4. **When the bounds bite** (Section 1.8). Identify the algebraic condition $\|\mu + s\| > \|\mu\|$ under which the bounds correspond to a genuine reduction relative to the unsteered baseline, discuss what happens when it fails (the anti-pole regime), and argue that it is generically satisfied in practice.

5. **Synthesis** (Section 1.9). Reconcile the local and global analyses, show they agree on scaling, and enumerate the testable empirical predictions.

6. **Experiments** (Section 2). Test all nine claims on Qwen2.5-1.5B-Instruct and Llama-3-8B-Instruct, with three trained EasySteer vectors and two random-direction controls. The headline finding: the scaling law is a statement about *aggregate steering magnitude*, not semantic direction; random vectors with matched aggregate magnitude reproduce trained-vector behaviour on Llama (slope $V = -1.85$ vs -1.92).

7. **Second-order consistency check** (Section 2.7). The parabolic shape of $V(s)$ on log–log axes is exactly the second-order prediction: $V(s) \propto 1/\|m(s)\|^2$ with $\|m(s)\|^2$ a quadratic in s. Fitting the quadratic recovers $\|\mu\|$ to within $\pm 10\%$ across all runs.

Proofs are interleaved with the statements throughout; readers focused on empirical claims may safely skim Sections 1–1.7 for the formulas in shaded boxes and proceed directly to Section 2. We refer to the nine quantitative predictions as "Claim 1" through "Claim 9"; the mapping between claim numbers and theorems is summarized in the empirical section's headline table.

1 A Diversity-Reduction Theory of Activation Steering

We develop a formal account of how constant additive steering vectors reduce the diversity of a transformer's internal representations. The core mechanism is the interaction between the fixed perturbation and the normalization layers (LayerNorm or RMSNorm) present in every transformer block: normalization projects out variation along the shifted mean and compresses the remaining variation by a factor that grows with the steering magnitude. We derive the output covariance of

a single normalization step under steering in closed form, characterize how effective rank and total variance depend on the steering magnitude, and discuss how the effect compounds across layers.

1.1 Setup and Notation

Let $x \in \mathbb{R}^d$ be a residual-stream vector drawn from a distribution $\mathcal{D}$ with mean $\mu = \mathbb{E}[x]$ and covariance $\Sigma_x = \mathbb{E}[(x-\mu)(x-\mu)^\top]$. A steering intervention adds a fixed vector $s \in \mathbb{R}^d$ to the residual stream, producing $x' = x + s$. This steered vector then passes through RMSNorm[1], which computes:

$$z = \frac{x+s}{\|x+s\|_{\mathrm{RMS}}}, \qquad \|v\|_{\mathrm{RMS}} = \sqrt{\frac{1}{d}\sum_{i=1}^{d} v_i^2} = \frac{\|v\|}{\sqrt{d}}. \tag{1}$$

Since $\|v\|_{\mathrm{RMS}} = \|v\|/\sqrt{d}$, RMSNorm is equivalent to projection onto the sphere of radius $\sqrt{d}$, followed by element-wise multiplication by learnable scale parameters γ. Because γ is fixed after training and does not depend on x, it acts as a constant linear transformation that does not affect rank. We therefore analyze the normalized vector up to the constant $\sqrt{d}$:

$$z = g(x+s), \qquad g(v) := \frac{v}{\|v\|}. \tag{2}$$

1.2 First-Order Expansion of the Normalization Map

The function $g\colon \mathbb{R}^d \setminus \{0\} \to S^{d-1}$ is nonlinear, so we linearize it around the shifted mean $m := \mu + s$ to obtain a tractable expression for the output covariance. Write $x + s = m + \delta$ where $\delta = x - \mu$ satisfies $\mathbb{E}[\delta] = 0$ and $\mathbb{E}[\delta\delta^\top] = \Sigma_x$.

Lemma 1.1 (Jacobian of the sphere projection)**.** *The Jacobian of $g(v) = v/\|v\|$ is*

$$J_g(v) = \frac{1}{\|v\|}\left(I - \hat{v}\hat{v}^\top\right), \tag{3}$$

where $\hat{v} = v/\|v\|$ is the unit vector along v.

Proof. We compute the (i,j)-entry of J_g directly. Write $g_i(v) = v_i/h(v)$ where $h(v) = \|v\| = (v^\top v)^{1/2}$. The partial derivative of h with respect to v_j is:

$$\frac{\partial h}{\partial v_j} = \frac{1}{2}\left(\sum_k v_k^2\right)^{-1/2} \cdot 2v_j = \frac{v_j}{\|v\|}. \tag{4}$$

By the quotient rule:

$$\frac{\partial g_i}{\partial v_j} = \frac{\delta_{ij}\cdot\|v\| - v_i \cdot \frac{v_j}{\|v\|}}{\|v\|^2} = \frac{\delta_{ij}}{\|v\|} - \frac{v_i v_j}{\|v\|^3}. \tag{5}$$

In matrix form, the first term is $\frac{1}{\|v\|}I$ and the second is $\frac{1}{\|v\|^3}vv^\top = \frac{1}{\|v\|}\hat{v}\hat{v}^\top$, giving (3). □

Applying the first-order Taylor expansion $g(m+\delta) \approx g(m) + J_g(m)\,\delta$:

$$z \approx \hat{m} + \frac{1}{\|m\|}\,P_{\perp m}\,\delta, \tag{6}$$

where $P_{\perp m} := I - \hat{m}\hat{m}^\top$ is the orthogonal projector onto the subspace perpendicular to m.

[1] We analyze RMSNorm rather than full LayerNorm for clarity. LayerNorm additionally subtracts the per-vector mean, which projects onto a $(d-1)$-dimensional subspace orthogonal to the all-ones vector $\mathbf{1}$ [**brody2023expressivity**]. This introduces a second rank-reducing projection but does not alter the core argument. In modern architectures (LLaMA, Qwen, etc.), RMSNorm has largely replaced LayerNorm.

Validity of the approximation. The expansion (6) is accurate when $\|\delta\| \ll \|m\|$, i.e., the input variation is small relative to the shifted mean. This condition is increasingly well-satisfied as the steering magnitude $\|s\|$ grows (since $\|m\| \geq \|s\| - \|\mu\|$[2]), which is precisely the regime where the diversity-reduction effect is strongest. The approximation breaks down when $s \approx -\mu$ so that $\|m\| \approx 0$; this anti-pole regime is analyzed in Section 1.8. Second-order corrections are of order $O(\|\delta\|^2/\|m\|^2)$.

1.3 Covariance Under Steering

From (6), the output has mean $\mathbb{E}[z] \approx \hat{m}$ and fluctuation $z - \hat{m} \approx \frac{1}{\|m\|} P_{\perp m}\, \delta$. The output covariance is therefore, using the fact that $P_{\perp m}$ is symmetric[3]:

$$\Sigma_z \;=\; \mathbb{E}\Big[(z-\hat{m})(z-\hat{m})^\top\Big] \;\approx\; \frac{1}{\|m\|^2}\, P_{\perp m}\, \Sigma_x\, P_{\perp m}^\top \;=\; \frac{1}{\|m\|^2}\, P_{\perp m}\, \Sigma_x\, P_{\perp m}. \tag{7}$$

Two effects are visible in (7):

1. **Rank reduction via projection.** The projector $P_{\perp m}$ has rank $d-1$: it annihilates all variation along $\hat{m}$. Geometrically, projecting onto the unit sphere eliminates radial variation—any component of δ along m changes only the norm $\|m+\delta\|$, not the direction $\widehat{m+\delta}$. The direction that is annihilated, $\hat{m} = \widehat{\mu+s}$, depends on s; different steering vectors eliminate different dimensions of variation.

2. **Variance shrinkage via rescaling.** All surviving variation is scaled by $1/\|m\|^2$. As $\|s\|$ grows, $\|m\| = \|\mu+s\|$ grows (at least as fast as $\|s\| - \|\mu\|$), compressing the total variance toward zero. This is the formalization of the informal observation by **[turntrout2023steering]** that when the steering coefficient is large, the steering vector "comprises most of the residual stream content" after LayerNorm.

Proposition 1.2 (Covariance under steered RMSNorm)**.** *Let $\mathcal{X}$ be a distribution over $\mathbb{R}^d$ with covariance Σ_x, let $s \in \mathbb{R}^d$ be a fixed steering vector, and let $m = \mathbb{E}[x] + s$. Under the first-order approximation* (6)*, the covariance Σ_z of the RMSNorm output satisfies:*

1. (Rank reduction) $\operatorname{rank}(\Sigma_z) \leq \min\big(\operatorname{rank}(\Sigma_x),\ d-1\big)$. *If Σ_x has nonzero variance along $\hat{m}$ (i.e., $\hat{m}^\top \Sigma_x \hat{m} > 0$), then* $\operatorname{rank}(\Sigma_z) \leq \operatorname{rank}(\Sigma_x) - 1$.

2. (Total variance reduction) $\operatorname{tr}(\Sigma_z) = \frac{1}{\|m\|^2}\big(\operatorname{tr}(\Sigma_x) - \hat{m}^\top \Sigma_x\, \hat{m}\big)$.

Proof. For the rank bound: $\Sigma_z = \frac{1}{\|m\|^2} P_{\perp m} \Sigma_x P_{\perp m}$, so $\operatorname{rank}(\Sigma_z) = \operatorname{rank}(P_{\perp m}\Sigma_x P_{\perp m})$. Since $P_{\perp m}$ has rank $d-1$, the product has rank at most $\min(\operatorname{rank}(\Sigma_x), d-1)$. If $\hat{m}^\top \Sigma_x \hat{m} > 0$, then Σ_x has a component in the nullspace of $P_{\perp m}$, and the rank drops by at least one.

[2] This is the reverse triangle inequality: $\|a+b\| \geq \big|\|a\| - \|b\|\big| \geq \|a\| - \|b\|$. One-line proof: write $s = m + (-\mu)$ and apply the ordinary triangle inequality, $\|s\| \leq \|m\| + \|\mu\|$, then rearrange. Geometrically, $m = \mu + s$ lies in a ball of radius $\|\mu\|$ centered at s, so the closest m can come to the origin is $\|s\| - \|\mu\|$ (achieved when $\mu \parallel -s$). The bound is only informative once $\|s\| > \|\mu\|$; otherwise $\|s\| - \|\mu\|$ is negative and the inequality is vacuous. We use this step in several places below (e.g. Section 1.3, and in the geometric-bound proofs of Section 1.6).

[3] $P_{\perp m}^\top = (I - \hat{m}\hat{m}^\top)^\top = I - \hat{m}\hat{m}^\top = P_{\perp m}$, because $\hat{m}\hat{m}^\top$ is an outer product of a vector with itself.

For the total variance: using the cyclic property of the trace, $\mathrm{tr}(ABC) = \mathrm{tr}(CAB)$, and the idempotence of $P_{\perp m}$:

$$\begin{aligned}
\mathrm{tr}(\Sigma_z) &= \frac{1}{\|m\|^2}\,\mathrm{tr}(P_{\perp m}\,\Sigma_x\,P_{\perp m}) && (8)\\
&= \frac{1}{\|m\|^2}\,\mathrm{tr}(P_{\perp m}^2\,\Sigma_x) && (9)\\
&= \frac{1}{\|m\|^2}\,\mathrm{tr}(P_{\perp m}\,\Sigma_x) && (10)\\
&= \frac{1}{\|m\|^2}\,\mathrm{tr}\big((I-\hat{m}\hat{m}^\top)\Sigma_x\big) && (11)\\
&= \frac{1}{\|m\|^2}\big(\mathrm{tr}(\Sigma_x)-\mathrm{tr}(\hat{m}\hat{m}^\top\Sigma_x)\big) && (12)\\
&= \frac{1}{\|m\|^2}\big(\mathrm{tr}(\Sigma_x)-\hat{m}^\top\Sigma_x\,\hat{m}\big). && (13)
\end{aligned}$$

The step from (8) to (9) uses the cyclic property to move the rightmost $P_{\perp m}$ to the left. The step from (9) to (10) uses idempotence: $P_{\perp m}^2 = P_{\perp m}$, which holds because $(I-\hat{m}\hat{m}^\top)^2 = I - 2\hat{m}\hat{m}^\top + \hat{m}(\hat{m}^\top\hat{m})\hat{m}^\top = I - \hat{m}\hat{m}^\top$, where $\hat{m}^\top\hat{m} = 1$ since $\hat{m}$ is a unit vector. The step from (12) to (13) uses the cyclic property to write $\mathrm{tr}(\hat{m}\hat{m}^\top\Sigma_x) = \mathrm{tr}(\hat{m}^\top\Sigma_x\hat{m}) = \hat{m}^\top\Sigma_x\hat{m}$, where the last equality holds because $\hat{m}^\top\Sigma_x\hat{m}$ is a scalar. □

Interpretation. The numerator $\mathrm{tr}(\Sigma_x)-\hat{m}^\top\Sigma_x\hat{m}$ is the original total variance minus the variance along the direction $\hat{m} = \widehat{\mu+s}$. For generic s, this quantity is strictly less than $\mathrm{tr}(\Sigma_x)$. The denominator $\|m\|^2 = \|\mu+s\|^2$ grows quadratically with the steering magnitude. In the strong-steering regime $\|s\| \gg \|\mu\|$:

$$\mathrm{tr}(\Sigma_z) \approx \frac{\mathrm{tr}(\Sigma_x) - \hat{s}^\top\Sigma_x\,\hat{s}}{\|s\|^2}, \tag{14}$$

which vanishes as $\|s\| \to \infty$.

1.4 Secondary Mechanisms

Softmax attention entropy. Steering shifts the query vectors by a constant, modifying the attention logits $q^\top k_i$. Because softmax is a concave function on the probability simplex with respect to uniform perturbations of the logits, a constant shift toward particular key positions reduces the entropy of the attention distribution. Lower attention entropy means fewer positions contribute to the output, further reducing the diversity of information flowing through the layer. We note this as a secondary contributor that is harder to formalize tightly.

Compounding across layers. In a pre-norm transformer, each block applies the sequence: Norm → Attn → Add → Norm → MLP → Add. The steering vector s, once added to the residual stream, persists via the residual connections. At each subsequent normalization step, the projection and shrinkage from Proposition 1.2 apply with the current shifted mean at that layer. In principle, effective rank can drop by up to one at each normalization step. However, the MLP and attention sub-layers are nonlinear maps that can expand variation (e.g., a ReLU/GeLU activation can map low-variance inputs to higher-variance outputs if the inputs straddle the nonlinearity). The net effect at each layer is therefore normalization-induced compression minus sub-layer-induced expansion. [**sinii2025small**] provide direct empirical evidence for the normalization bottleneck:

they find that steering vectors placed before the input LayerNorm of a block lose effectiveness, and that placing the vector *after* the LayerNorm restores performance—indicating that the LayerNorm step specifically attenuates the steering signal.

From representation diversity to output diversity. The unembedding layer is a linear map $W_U \in \mathbb{R}^{V \times d}$ (where V is vocabulary size) that converts residual-stream vectors to logits. The covariance of the logit distribution is $W_U \Sigma_z W_U^\top$, whose rank is at most $\mathrm{rank}(\Sigma_z)$. Since rank cannot increase through linear maps, representation-level diversity reduction is an *upper bound* on output-level diversity, which is the direction needed for our claims.

1.5 Relationship to Prior Work

[**dong2021attention**] showed that pure self-attention without skip connections suffers from doubly exponential rank collapse with depth. [**wu2024role**] extended this analysis to include LayerNorm, showing that LayerNorm can either prevent or accelerate collapse depending on the value matrices—importantly establishing that the rank dynamics of self-attention with LayerNorm are richer than previously supposed. Our analysis differs from this line of work in that we study rank reduction caused not by depth but by an *external additive perturbation* interacting with normalization. The depth-collapse results concern the unsteered model's intrinsic dynamics; our result concerns a controlled intervention.

[**turntrout2023steering**] informally observed that GPT-2's LayerNorm renormalizes the residual stream to $\sqrt{d_{\text{model}}}$, so that a large steering coefficient causes the steering vector to dominate the post-norm content. Our Proposition 1.2 formalizes this observation: the "domination" is precisely the variance shrinkage by $1/\|m\|^2$, and the "most of the residual stream content" corresponds to the projection $P_{\perp m}$ annihilating variation along the steering direction. The companion post on residual stream norm growth [**turntrout2023norms**] establishes that model weights compensate for LayerNorm's normalization by producing exponentially growing outputs—a mechanism that steering vectors, being fixed and not adapted to the model's learned growth schedule, do not participate in.

Most directly, [**sinii2025small**] empirically demonstrate that the input LayerNorm is the specific bottleneck limiting steering vector effectiveness between adjacent layers, and that bypassing it recovers performance. They frame this as an engineering obstacle; we reframe it as the *primary architectural mechanism* through which steering compresses representational diversity.

On the output side, [**zhang2025verbalized**] attribute post-alignment mode collapse to typicality bias in preference data, and [**omahony2024attributing**] attribute it to the mode-seeking nature of reverse KL in RLHF/DPO. These are complementary explanations at the level of training signal; our account provides a mechanistic explanation at the level of architecture for why inference-time steering interventions produce analogous diversity reduction without any training.

1.6 A Global Geometric Bound

The first-order analysis of Section 1.3 characterizes the output covariance structure but relies on a local linear approximation. We now give a complementary global result that holds exactly for arbitrary input distributions, using only elementary properties of the normalization map.

1.6.1 Chordal Contraction Under Normalization

The normalization map $\phi(v) = v/\|v\|$ projects $\mathbb{R}^d \setminus \{0\}$ onto the unit sphere S^{d-1}. The following lemma bounds how far apart two points can land on the sphere as a function of their Euclidean separation and magnitudes.

Lemma 1.3 (Chordal contraction)**.** *For any nonzero $u, v \in \mathbb{R}^d$:*

$$\|\hat{u} - \hat{v}\| \;\leq\; \frac{2\,\|u - v\|}{\max(\|u\|, \|v\|)}. \tag{15}$$

Proof. Without loss of generality, assume $\|u\| \geq \|v\|$ (the bound is symmetric under swapping u and v). Decompose:

$$\hat{u} - \hat{v} = \frac{u}{\|u\|} - \frac{v}{\|v\|} \;=\; \frac{u - v}{\|u\|} + v\left(\frac{1}{\|u\|} - \frac{1}{\|v\|}\right) \tag{16}$$

$$= \frac{u - v}{\|u\|} + \frac{v\,(\|v\| - \|u\|)}{\|u\|\,\|v\|}. \tag{17}$$

Taking norms and applying the triangle inequality:

$$\|\hat{u} - \hat{v}\| \;\leq\; \frac{\|u - v\|}{\|u\|} + \frac{\|v\| \cdot \big|\|v\| - \|u\|\big|}{\|u\|\,\|v\|} \;=\; \frac{\|u - v\|}{\|u\|} + \frac{\big|\|v\| - \|u\|\big|}{\|u\|}. \tag{18}$$

By the reverse triangle inequality, $\big|\|v\| - \|u\|\big| \leq \|u - v\|$, so:

$$\|\hat{u} - \hat{v}\| \;\leq\; \frac{2\,\|u - v\|}{\|u\|} \;=\; \frac{2\,\|u - v\|}{\max(\|u\|, \|v\|)}. \tag{19}$$

□

1.6.2 Concentration Around the Steering Pole

Let $\phi_s(x) = \widehat{x + s}$ denote the normalized steered representation. We first bound how far any single point can land from the "pole" $\hat{s}$.

Proposition 1.4 (Concentration around $\hat{s}$)**.** *For any $x \in \mathbb{R}^d$ and any nonzero $s \in \mathbb{R}^d$:*

$$\|\phi_s(x) - \hat{s}\| \;\leq\; \frac{2\,\|x\|}{\|s\|}. \tag{20}$$

Proof. Apply Lemma 1.3 with $u = x + s$ and $v = s$:

$$\|\phi_s(x) - \hat{s}\| = \|\widehat{x + s} - \hat{s}\| \leq \frac{2\,\|(x + s) - s\|}{\max(\|x + s\|, \|s\|)} = \frac{2\,\|x\|}{\max(\|x + s\|, \|s\|)}. \tag{21}$$

Since $\max(\|x + s\|, \|s\|) \geq \|s\|$, the result follows. □

This is the formalization of the geometric intuition: every point on the sphere is drawn toward the pole $\hat{s}$, with the maximum displacement from the pole shrinking as $O(1/\|s\|)$. Critically, this bound is *exact*—it involves no Taylor expansion or distributional assumptions.

Remark 1.5 (The anti-pole)**.** *If $x \approx -s$, then $x + s \approx 0$ and $\phi_s(x)$ is ill-defined or can point in an arbitrary direction. However, for any distribution $\mathcal{D}$ with bounded support $\|x\| \leq R < \|s\|$, this regime is inaccessible: $\|x+s\| \geq \|s\| - R > 0$ for all x in the support. A subtler issue arises when the* mean *satisfies $\mu \approx -s$, so that the shifted distribution straddles the origin even though no individual point reaches it. In this regime, normalization amplifies rather than compresses variation. This case is analyzed in Section 1.8.*

1.6.3 Pairwise Contraction

Proposition 1.6 (Pairwise chordal bound)**.** *For any $x_1, x_2 \in \mathbb{R}^d$ and nonzero s with $\|s\| > R :=$* $\max(\|x_1\|, \|x_2\|)$*:*

$$\|\phi_s(x_1) - \phi_s(x_2)\| \;\leq\; \frac{2\,\|x_1 - x_2\|}{\|s\| - R}. \tag{22}$$

Proof. Apply Lemma 1.3 with $u = x_1 + s$ and $v = x_2 + s$. Since $(x_1 + s) - (x_2 + s) = x_1 - x_2$, the steering vector cancels in the numerator:

$$\|\phi_s(x_1) - \phi_s(x_2)\| \;\leq\; \frac{2\,\|x_1 - x_2\|}{\max(\|x_1 + s\|, \|x_2 + s\|)}. \tag{23}$$

For the denominator: by the triangle inequality, $\|x_i + s\| \geq \|s\| - \|x_i\| \geq \|s\| - R$ for each i, so $\max(\|x_1 + s\|, \|x_2 + s\|) \geq \|s\| - R$. □

The numerator $\|x_1 - x_2\|$ is the Euclidean distance between the *unsteered* inputs—the steering cancels. The denominator grows linearly with $\|s\|$. Thus, the map ϕ_s is Lipschitz on the support of the distribution with constant $L = 2/(\|s\| - R)$, which becomes a strict contraction ($L < 1$) when $\|s\| > R + 2$.

Remark 1.7 (Steering can increase individual pairwise angles)**.** *The bound* (22) *constrains the* absolute *post-steering angular spread, but does not claim that steering reduces the angular distance for every pair relative to the unsteered baseline. A counterexample: in $\mathbb{R}^2$, take $x_1 = (1, 0)$, $x_2 = (0, 1)$, and $s = (-\frac{1}{2}, 0)$. The unsteered angle between $\hat{x}_1$ and $\hat{x}_2$ is $\pi/2$, but the post-steering angle is* $\arccos(-1/\sqrt{5}) \approx 2.03$ *radians—an increase. This occurs because s moves x_1 toward the origin, amplifying angular sensitivity in that region. The global bound still holds: as $\|s\|$ grows, such effects vanish because all points are pulled away from the origin toward the pole.*

1.6.4 Distribution-Level Diversity Bounds

Let $\mathcal{D}$ be a distribution over $\mathbb{R}^d$ with bounded support $\|x\| \leq R$ and let $\phi_s(\mathcal{D})$ denote the pushforward distribution on S^{d-1}.

Corollary 1.8 (Spherical variance bound)**.** *The spherical variance $V(\phi_s(\mathcal{D})) := 1 - \|\bar{R}\|$ where $\bar{R} = \mathbb{E}_{x\sim\mathcal{D}}[\phi_s(x)]$ is the mean resultant vector* ***[mardia2000directional]****, satisfies:*

$$V(\phi_s(\mathcal{D})) \;\leq\; \frac{2\,\mathbb{E}[\|x\|]}{\|s\|}. \tag{24}$$

Proof. By Proposition 1.4, $\|\phi_s(x) - \hat{s}\| \leq 2\|x\|/\|s\|$ for each x. Taking expectations:

$$\|\bar{R} - \hat{s}\| = \|\mathbb{E}[\phi_s(x)] - \hat{s}\| = \|\mathbb{E}[\phi_s(x) - \hat{s}]\| \leq \mathbb{E}[\|\phi_s(x) - \hat{s}\|] \leq \frac{2\,\mathbb{E}[\|x\|]}{\|s\|}. \tag{25}$$

Since $\|\hat{s}\| = 1$, the reverse triangle inequality gives $\|\bar{R}\| \geq 1 - \|\bar{R} - \hat{s}\|$, so:

$$V = 1 - \|\bar{R}\| \leq \|\bar{R} - \hat{s}\| \leq \frac{2\,\mathbb{E}[\|x\|]}{\|s\|}. \tag{26}$$

□

Corollary 1.9 (Expected pairwise chordal distance)**.** *For independent $x_1, x_2 \sim \mathcal{D}$:*

$$\mathbb{E}[\|\phi_s(x_1) - \phi_s(x_2)\|] \;\leq\; \frac{4\,\mathbb{E}[\|x\|]}{\|s\|}. \tag{27}$$

Proof. By the triangle inequality through the pole: $\|\phi_s(x_1) - \phi_s(x_2)\| \leq \|\phi_s(x_1) - \hat{s}\| + \|\hat{s} - \phi_s(x_2)\|$. Applying Proposition 1.4 to each term and taking expectations over independent x_1, x_2:

$$\mathbb{E}[\|\phi_s(x_1) - \phi_s(x_2)\|] \leq \frac{2\,\mathbb{E}[\|x_1\|]}{\|s\|} + \frac{2\,\mathbb{E}[\|x_2\|]}{\|s\|} = \frac{4\,\mathbb{E}[\|x\|]}{\|s\|}. \tag{28}$$

□

Corollary 1.10 (Spherical diameter)**.** *The diameter of $\phi_s(\mathcal{D})$ in chordal distance satisfies:*

$$\operatorname{diam}(\phi_s(\mathcal{D})) \;\leq\; \frac{4R}{\|s\|}, \tag{29}$$

where $R = \sup\{\|x\| : x \in \operatorname{supp}(\mathcal{D})\}$.

Proof. For any x_1, x_2 in the support, Proposition 1.4 gives $\|\phi_s(x_i) - \hat{s}\| \leq 2R/\|s\|$. By the triangle inequality, $\|\phi_s(x_1) - \phi_s(x_2)\| \leq 4R/\|s\|$. □

1.6.5 Interpretation

These results formalize the geometric picture of diversity reduction under steering. The distribution of post-normalization representations is confined to a spherical cap of chordal radius $2R/\|s\|$ centered at the pole $\hat{s}$. As the steering magnitude increases, this cap shrinks, and all measures of angular dispersion—spherical variance, expected pairwise distance, diameter—decrease as $O(1/\|s\|)$.

Comparison with the local analysis. The first-order expansion of Section 1.3 and the global bounds of this section are complementary:

	Local (Taylor)	**Global (geometric)**
Approximation	First-order; requires $\|\delta\| \ll \|m\|$	Exact
Diversity measure	Effective rank, eigenvalues of Σ_z	Spherical variance, chordal distance
Structural detail	Full covariance structure	Scalar bounds only
Applicable regime	Large $\|s\|$ (increasingly accurate)	All $\|s\| > 0$

The local analysis reveals *which* dimensions of variation are lost (those along $\hat{m}$) and by how much (the eigenvalues of $P_{\perp m}\Sigma_x P_{\perp m}/\|m\|^2$). The global analysis establishes that post-normalization spread is *bounded* for any steering magnitude, without approximation error, and gives quantitative scaling with $\|s\|$. Whether this bound implies a *reduction* relative to the unsteered baseline depends on additional conditions analyzed in Section 1.8.

1.7 Second-Order Corrections

The first-order expansion of Section 1.3 gives the leading behavior but leaves open the question of how large the approximation error is. We now compute the second-order correction, which reveals additional structure and confirms consistency with the global geometric bounds of Section 1.6.

The Hessian of the sphere projection. We require the second derivatives of $g_i(v) = v_i/\|v\|$. Starting from the Jacobian entry derived in Lemma 1.1:

$$\frac{\partial g_i}{\partial v_j} = \frac{\delta_{ij}}{\|v\|} - \frac{v_i v_j}{\|v\|^3}, \tag{30}$$

we differentiate with respect to v_k. The first term contributes $-\delta_{ij}\, v_k/\|v\|^3$ via $\partial_{v_k}(\|v\|^{-1}) = -v_k/\|v\|^3$. The second term contributes, by the product and chain rules:

$$-\frac{\delta_{ik}\, v_j + v_i\, \delta_{jk}}{\|v\|^3} + \frac{3\, v_i\, v_j\, v_k}{\|v\|^5}. \tag{31}$$

Combining:

$$\frac{\partial^2 g_i}{\partial v_j \partial v_k} = -\frac{\delta_{ij}\, v_k + \delta_{ik}\, v_j + v_i\, \delta_{jk}}{\|v\|^3} + \frac{3\, v_i\, v_j\, v_k}{\|v\|^5}. \tag{32}$$

Contracting with δ. The second-order term in the Taylor expansion is $\frac{1}{2}\sum_{j,k} \frac{\partial^2 g_i}{\partial v_j \partial v_k}\big|_{v=m} \delta_j\, \delta_k$. Evaluating (32) at $v = m$ and summing over j and k, the four groups yield:

$$-\frac{\delta_{ij}\, m_k\, \delta_j\, \delta_k}{\|m\|^3} = -\frac{\delta_i\, (\hat{m}^\top \delta)}{\|m\|^2}, \tag{33}$$

$$-\frac{\delta_{ik}\, m_j\, \delta_j\, \delta_k}{\|m\|^3} = -\frac{(\hat{m}^\top \delta)\, \delta_i}{\|m\|^2}, \tag{34}$$

$$-\frac{m_i\, \delta_{jk}\, \delta_j\, \delta_k}{\|m\|^3} = -\frac{\hat{m}_i\, \|\delta\|^2}{\|m\|^2}, \tag{35}$$

$$\frac{3\, m_i\, m_j\, m_k\, \delta_j\, \delta_k}{\|m\|^5} = \frac{3\, \hat{m}_i\, (\hat{m}^\top \delta)^2}{\|m\|^2}. \tag{36}$$

Summing and writing in vector form:

$$H_g(m)[\delta, \delta] = \frac{1}{\|m\|^2}\Big[-2(\hat{m}^\top \delta)\, \delta - \|\delta\|^2\, \hat{m} + 3(\hat{m}^\top \delta)^2\, \hat{m}\Big]. \tag{37}$$

Decomposition into radial and perpendicular components. Let $\alpha = \hat{m}^\top \delta$ denote the radial component of δ (its projection onto $\hat{m}$), and let $\delta_\perp = P_{\perp m}\, \delta = \delta - \alpha\, \hat{m}$ denote the perpendicular component. Then $\|\delta\|^2 = \alpha^2 + \|\delta_\perp\|^2$, and substituting into (37):

$$\begin{aligned} H_g(m)[\delta, \delta] &= \frac{1}{\|m\|^2}\Big[-2\alpha\, (\alpha\, \hat{m} + \delta_\perp) - (\alpha^2 + \|\delta_\perp\|^2)\, \hat{m} + 3\alpha^2\, \hat{m}\Big] \\ &= \frac{1}{\|m\|^2}\Big[\underbrace{(-2\alpha^2 - \alpha^2 - \|\delta_\perp\|^2 + 3\alpha^2)}_{= -\|\delta_\perp\|^2}\, \hat{m} - 2\alpha\, \delta_\perp\Big] \\ &= \frac{1}{\|m\|^2}\Big[-\|\delta_\perp\|^2\, \hat{m} - 2\alpha\, \delta_\perp\Big]. \end{aligned} \tag{38}$$

The α^2 terms in the $\hat{m}$ direction cancel exactly.

Full second-order expansion. Combining the first-order term from (6) with $\frac{1}{2}H_g(m)[\delta,\delta]$:

$$z \approx \hat{m} + \frac{1}{\|m\|}\,\delta_\perp + \frac{1}{2\|m\|^2}\Big[-\|\delta_\perp\|^2\,\hat{m} - 2\alpha\,\delta_\perp\Big] + O\!\left(\frac{\|\delta\|^3}{\|m\|^3}\right). \tag{39}$$

Regrouping by direction:

$$\boxed{z \approx \left(1 - \frac{\|\delta_\perp\|^2}{2\|m\|^2}\right)\hat{m} \;+\; \left(\frac{1}{\|m\|} - \frac{\alpha}{\|m\|^2}\right)\delta_\perp \;+\; O\!\left(\frac{\|\delta\|^3}{\|m\|^3}\right).} \tag{40}$$

Two effects appear at second order:

1. **Radial pullback.** The coefficient of $\hat{m}$ decreases from 1 by $\|\delta_\perp\|^2/(2\|m\|^2)$. This is the sphere curvature: spreading perpendicular to the pole pulls the projection inward, reducing the mean resultant length. This is precisely the mechanism underlying the spherical variance bound of Corollary 1.8.

2. **Radial-perpendicular coupling.** The coefficient of $\delta_\perp$ acquires a correction $-\alpha/\|m\|^2$ that depends on the radial fluctuation $\alpha = \hat{m}^\top\delta$. Points with $\alpha > 0$ (further from the origin) receive *less* angular spread; points with $\alpha < 0$ (closer to the origin) receive *more*. This is the perturbative manifestation of the magnifying-glass effect discussed in Section 1.8.2: the normalization map amplifies angular variation near the origin.

Second-order mean. Taking expectations using $\mathbb{E}[\delta] = 0$ (and therefore $\mathbb{E}[\alpha] = 0$, $\mathbb{E}[\delta_\perp] = 0$):

$$\mathbb{E}[z] \approx \hat{m} - \frac{\mathbb{E}[\|\delta_\perp\|^2]}{2\|m\|^2}\,\hat{m} - \frac{\mathbb{E}[\alpha\,\delta_\perp]}{\|m\|^2}. \tag{41}$$

For the first correction: $\mathbb{E}[\|\delta_\perp\|^2] = \mathrm{tr}(P_{\perp m}\,\Sigma_x\,P_{\perp m}) = \mathrm{tr}(\Sigma_x) - \hat{m}^\top\Sigma_x\,\hat{m}$, which is the perpendicular variance from Proposition 1.2. For the second correction: $\mathbb{E}[\alpha\,\delta_\perp] = \mathbb{E}[(\hat{m}^\top\delta)\,P_{\perp m}\,\delta] = P_{\perp m}\,\Sigma_x\,\hat{m}$, which is the cross-covariance between the radial and perpendicular components of δ. This vanishes when $\hat{m}$ is an eigenvector of Σ_x or when Σ_x is isotropic.

The mean resultant length is therefore:

$$\|\mathbb{E}[z]\| \approx 1 - \frac{\mathrm{tr}(\Sigma_x) - \hat{m}^\top\Sigma_x\,\hat{m}}{2\|m\|^2} + O\!\left(\frac{\|\Sigma_x\|^2}{\|m\|^4}\right), \tag{42}$$

where the cross-covariance term contributes only at $O(1/\|m\|^4)$ to the norm (since it is perpendicular to the leading $\hat{m}$ term). The spherical variance is thus:

$$V := 1 - \|\mathbb{E}[z]\| \approx \frac{\mathrm{tr}(\Sigma_x) - \hat{m}^\top\Sigma_x\,\hat{m}}{2\|m\|^2}, \tag{43}$$

which scales as $O(1/\|m\|^2) = O(1/\|s\|^2)$ in the strong-steering regime, consistent with the $O(1/\|s\|)$ chordal bound from Corollary 1.8 (since spherical variance scales as the square of the chordal spread for concentrated distributions).

Second-order covariance. The fluctuation around the corrected mean is, to second order:

$$\varepsilon = z - \mathbb{E}[z] = \frac{1}{\|m\|}\,\delta_\perp - \frac{1}{\|m\|^2}\Big[\alpha\,\delta_\perp - \mathbb{E}[\alpha\,\delta_\perp]\Big] - \frac{1}{2\|m\|^2}\Big[\|\delta_\perp\|^2 - \mathbb{E}[\|\delta_\perp\|^2]\Big]\,\hat{m} + O\left(\frac{\|\delta\|^3}{\|m\|^3}\right). \quad (44)$$

The covariance $\Sigma_z = \mathbb{E}[\varepsilon\,\varepsilon^\top]$ has three tiers:

1. **Leading term**, $O(1/\|m\|^2)$: $\frac{1}{\|m\|^2}\,P_{\perp m}\,\Sigma_x\,P_{\perp m}$, exactly as in Proposition 1.2.
2. **Cross terms**, $O(1/\|m\|^3)$: these involve third central moments of δ, specifically $\mathbb{E}[\delta_\perp \cdot \alpha\,\delta_\perp^\top]$ and its transpose. *These vanish for any distribution symmetric about μ.* In particular, for Gaussian inputs the first-order covariance is accurate to $O(1/\|m\|^4)$, not merely $O(1/\|m\|^3)$.
3. **Quadratic terms**, $O(1/\|m\|^4)$: these involve fourth central moments such as $\mathbb{E}[\alpha^2\,\delta_\perp\,\delta_\perp^\top]$ and $\mathbb{E}[\|\delta_\perp\|^4]$. For Gaussian inputs these can be expressed via Isserlis' theorem in terms of Σ_x, but the resulting expressions are cumbersome and we omit them.

Summary. The second-order analysis confirms and sharpens the first-order results:

Quantity	**First-order**	**Second-order correction**
Mean direction	$\hat{m}$	unchanged
Mean resultant length	1	$1 - \dfrac{\mathrm{tr}(\Sigma_x) - \hat{m}^\top\Sigma_x\hat{m}}{2\|m\|^2}$
Spherical variance	0 (trivially)	$\dfrac{\mathrm{tr}(\Sigma_x) - \hat{m}^\top\Sigma_x\hat{m}}{2\|m\|^2}$
Covariance	$\dfrac{P_{\perp m}\,\Sigma_x\,P_{\perp m}}{\|m\|^2}$	same $+O(1/\|m\|^4)$ for symmetric δ

The key takeaway is that for symmetric input distributions (including Gaussian), the first-order covariance formula from Proposition 1.2 is more accurate than its derivation might suggest: the next correction is $O(1/\|m\|^4)$, not $O(1/\|m\|^3)$. For asymmetric distributions, third-moment effects enter at $O(1/\|m\|^3)$ and can either increase or decrease the covariance depending on the skewness structure.

1.8 When Does Steering Reduce Diversity?

The results of Sections 1.3 and 1.6 establish that the post-normalization distribution concentrates around the pole $\hat{s}$ with spread shrinking as $O(1/\|s\|)$. However, neither result directly compares the steered diversity to the *unsteered* baseline. In this section, we identify the condition under which steering genuinely reduces diversity relative to the unsteered model, show that the condition can fail, and argue that it is satisfied in all practical scenarios.

1.8.1 The Condition

Consider the unsteered case ($s = 0$). The normalization map sends x to $\hat{x} = x/\|x\|$, projecting the distribution $\mathcal{D}$ onto the sphere. The total variance of the unsteered output is governed (via the first-order analysis) by $1/\|\mu\|^2$, and the steered output by $1/\|\mu + s\|^2$. Steered diversity is lower than unsteered diversity when the denominator grows, i.e., when:

$$\|\mu + s\| > \|\mu\|. \quad (45)$$

Squaring both sides and expanding:

$$\|\mu\|^2 + 2\mu^\top s + \|s\|^2 > \|\mu\|^2, \tag{46}$$

which simplifies to:

$$2\mu^\top s + \|s\|^2 > 0 \qquad \Longleftrightarrow \qquad \cos\angle(\mu, s) > -\frac{\|s\|}{2\|\mu\|}. \tag{47}$$

This is a weak requirement. For any $\|s\| > 0$, the right-hand side is negative, so the condition is satisfied whenever $\cos\angle(\mu, s) \geq 0$ (i.e., the steering vector is in the same half-space as the mean), and even permits obtuse angles up to $\arccos(-\|s\|/2\|\mu\|)$. Only when the steering vector is nearly anti-parallel to the mean and small relative to it can the condition fail.

1.8.2 Failure at the Anti-Pole

When $s \approx -\mu$, the shifted mean $m = \mu + s$ is near the origin. In this regime:

- The first-order approximation breaks down: the condition $\|\delta\| \ll \|m\|$ is violated, and the predicted variance $\mathrm{tr}(\Sigma_x)/\|m\|^2$ diverges.
- Geometrically, the distribution of shifted vectors $\{x + s\}$ straddles the origin. Points on opposite sides of the origin are projected to opposite sides of the sphere, so the normalization map acts as a *magnifying glass*: small Euclidean differences become large angular differences.
- The global concentration bound (Proposition 1.4) becomes vacuous: when $\|s\| \approx \|\mu\|$, the bound $2\,\mathbb{E}[\|x\|]/\|s\| \approx 2$ saturates the maximum possible chordal distance.

This is not merely a limitation of our proof technique. The diversity genuinely increases in this regime: the normalization map near the origin has unbounded Lipschitz constant (the Jacobian $J_g(v) = (I - \hat{v}\hat{v}^\top)/\|v\|$ has operator norm $1/\|v\|$), so it amplifies input variation.

1.8.3 Why the Condition Is Generically Satisfied

We argue that the anti-pole regime is not encountered in practice, for two independent reasons.

Geometric argument in high dimensions. In $\mathbb{R}^d$ with large d, two structured vectors are unlikely to be nearly anti-parallel. Two independent random unit vectors have cosine similarity with mean 0 and standard deviation $1/\sqrt{d}$; for $d = 4096$, achieving $\cos\angle(\mu, s) < -1/64$ (which is still far from the threshold $-\|s\|/2\|\mu\|$) would be a multi-sigma event. While μ and s are not random—they are structured vectors reflecting the model's learned representations—the same concentration of measure makes near-anti-alignment a priori unlikely without deliberate adversarial construction.

Mechanistic argument. Steering vectors are typically computed as activation *differences* between prompts exhibiting contrasting behaviors (e.g., eval-aware minus eval-unaware). These differences encode a behavioral contrast direction. The mean residual stream μ at a given layer encodes average content: positional information, syntactic structure, and semantic features aggregated over the input distribution. There is no mechanistic reason for a behavioral contrast direction to point toward the negation of the average content. To construct $s \approx -\mu$, one would need a steering vector that cancels the aggregate content of the residual stream—a pathological intervention that would destroy all model computation, not just reduce diversity.

Norm growth. [**turntrout2023norms**] document that residual stream norms grow exponentially with layer depth (approximately 1.045× per layer in GPT-2-XL). Steering vectors, being fixed additive perturbations, do not participate in this learned growth schedule. At deeper layers, $\|\mu\|$ is therefore typically much larger than $\|s\|$, making the condition (47) even easier to satisfy: the threshold $-\|s\|/2\|\mu\|$ approaches zero, and essentially any non-adversarial steering direction suffices.

1.8.4 Recommended Treatment

For the formal results, we state the condition $\|\mu + s\| > \|\mu\|$ as an explicit assumption. For all experimental results, we verify it empirically by computing $\|\mu + s\|/\|\mu\|$ at the injection layer and confirming the ratio exceeds 1. This is a lightweight check (requiring only the mean residual stream norm, which is already available from the evaluation pipeline) and provides a per-experiment certificate that the diversity-reduction theory applies.

1.9 Synthesis

We now summarize the full theoretical picture and clarify the relationship between the local and global analyses.

1.9.1 Scaling Correspondence

The global analysis (Section 1.6) gives bounds that shrink as $O(1/\|s\|)$: spherical variance, expected pairwise chordal distance, and diameter all scale inversely with the steering magnitude. The local analysis (Section 1.3) gives total variance $\mathrm{tr}(\Sigma_z) = O(1/\|s\|^2)$. These are consistent, not contradictory.

For a distribution concentrated near a point on the sphere, chordal distance c and angular distance θ are related by $c = 2\sin(\theta/2) \approx \theta$ for small θ. The variance of a scalar angular displacement scales as θ^2, while the angular displacement itself scales as θ. So $O(1/\|s\|)$ in chordal distance corresponds to $O(1/\|s\|^2)$ in variance, which is exactly the relationship between our two bounds.

More precisely, for a distribution on S^{d-1} concentrated near a pole with small angular spread θ, the spherical variance $V = 1 - \|\bar{R}\| \approx \theta^2/2$ and the trace of the tangent-space covariance $\mathrm{tr}(\Sigma_z) \approx \theta^2$. The global bound gives $\theta = O(1/\|s\|)$; squaring recovers the local bound.

1.9.2 What Each Result Provides

The two analyses are complementary, and each answers questions the other cannot:

The local (Taylor) analysis reveals the internal structure of the diversity reduction. It identifies *which* dimension of variation is lost (the direction $\hat{m} = \widehat{\mu + s}$), gives the full eigenvalue structure of the output covariance, and shows that effective rank drops by exactly one when Σ_x has nonzero variance along $\hat{m}$. This structural detail connects directly to the pass@k story: the lost dimension corresponds to a specific mode of computational variation that the model can no longer express. The cost is that the result is approximate, valid only when input variation is small relative to $\|\mu + s\|$.

The global (geometric) analysis provides exact, assumption-free bounds on the absolute spread of post-normalization representations. It requires no distributional assumptions beyond bounded support, and the bounds hold for any steering magnitude. The cost is that it provides

only scalar bounds (diameter, spherical variance), not structural information about which dimensions are compressed.

The condition analysis (Section 1.8) identifies when the diversity reduction is genuine relative to the unsteered baseline: $\|\mu + s\| > \|\mu\|$. This is the load-bearing assumption that connects the absolute concentration bounds to relative diversity claims.

1.9.3 The Complete Argument

Combining all three components, the complete theoretical argument is:

1. **Architecture.** Every transformer block contains normalization layers (RMSNorm or LayerNorm) that project the residual stream onto a sphere (or affine image thereof).
2. **Mechanism.** Adding a fixed steering vector s shifts the pre-normalization distribution. The normalization map then concentrates all representations toward the pole $\hat{s}$, with chordal spread bounded by $O(\mathbb{E}[\|x\|]/\|s\|)$ (Proposition 1.4).
3. **Structure.** The concentration specifically eliminates variation along the direction $\hat{m} = \widehat{\mu + s}$ and compresses all remaining variation by $1/\|\mu + s\|^2$ (Proposition 1.2).
4. **Condition.** This constitutes a *reduction* relative to the unsteered baseline when $\|\mu+s\| > \|\mu\|$ (Section 1.8), a condition satisfied generically in practice.
5. **Compounding.** The steering vector persists via residual connections, so the compression applies at every subsequent normalization layer. The net effect per layer is normalization-induced compression minus sub-layer-induced expansion; empirical evidence [**sinii2025small**] confirms that the normalization step is the specific bottleneck limiting steering vector effectiveness.
6. **Output.** Representational diversity upper-bounds output diversity, since the unembedding is a linear map that cannot increase rank.

Steps 1–3 are proved. Step 4 is a stated condition with strong practical justification. Step 5 is a qualitative argument supported by empirical evidence. Step 6 is exact.

1.9.4 Empirical Predictions

The theory makes several testable predictions:

1. **Effective rank decreases with steering magnitude.** Measuring the effective rank of the representation covariance at each layer, steered models should show lower effective rank than unsteered models, with the reduction scaling as $O(1/\|s\|^2)$ in variance terms.
2. **The lost dimension aligns with the steering direction.** The principal component of variance lost under steering should have high cosine similarity with $\hat{m} = \widehat{\mu + s}$.
3. **Diversity reduction concentrates at normalization layers.** Measuring representational diversity before and after each sub-component within a transformer block (following the methodology of [**sinii2025small**]), the largest drops should occur at the normalization steps, not at the attention or MLP steps.

4. **Pass@k degrades faster than pass@1.** If steering compresses the output distribution while preserving the mode, then pass@1 (which depends only on the mode) should be relatively preserved while pass@k for large k (which depends on distributional coverage) should degrade. This prediction connects the representation-level theory to the behavioral-level measurements in the experimental sections.

2 Empirical Verification of the Diversity-Reduction Bounds

We test the nine claims of Sections 1, 1.6, 1.8, and 1.9 empirically on two open-weight decoder-only transformers, using steering vectors drawn from the public EasySteer library and a broad web-text distribution. The claims divide naturally into three groups: (i) exact global bounds whose validity is a geometric identity independent of the data (Propositions 1.4 and 1.6 and Corollary 1.10); (ii) approximate local or asymptotic predictions whose tightness depends on the relationship between the steering magnitude and the residual-stream scale (Proposition 1.2, Corollaries 1.8 and 1.9, and the scaling law of Section 1.9); and (iii) conditions that determine whether the mechanism is even operating, such as the reduction gate of (45). All three classes are tested against the same empirical scaffolding: run a large number of broad natural-language prompts through the model, intervene on the residual stream at a chosen set of layers, and compute streaming statistics at the final RMSNorm site.

Full source code, configuration files, per-run output, and a master orchestration script for the full protocol are published at `https://github.com/AMindToThink/steering_diversity` under `src/bounds/`, `scripts/bounds/`, and `outputs/bounds/`. Every number reported in this section is reproducible by re-running `scripts/bounds/experiment2/run.sh`.

2.1 Setup

Models. We use two HuggingFace Transformers checkpoints spanning an order of magnitude in parameter count and architecture-dependent residual-stream norms:

- **Qwen2.5-1.5B-Instruct**: hidden size $d = 1536$, $L = 28$ decoder blocks, RMSNorm, RoPE.
- **Meta-Llama-3-8B-Instruct**: hidden size $d = 4096$, $L = 32$ decoder blocks, RMSNorm, RoPE.

Both architectures use Rotary Position Embedding applied inside attention (to query and key projections, not additively to the residual stream), so the residual stream never has a positional-embedding term added to it. No additive positional embedding appears in any of the empirical LHS or RHS quantities reported below.

Data. We use `HuggingFaceFW/fineweb-edu`, a broad, quality-filtered web-text corpus. Each run streams 1000 documents from the `train` split, truncates each to the first 256 tokens after tokenization, and applies one forward pass per document. Token-level activations (masked for padding) are the statistical unit. The choice of dataset is deliberate: we want a broad input distribution that stresses rather than cherry-picks the bounds, and `fineweb-edu` is among the broadest and most quality-controlled public pretraining corpora available as of 2025.

Steering vectors. We use three publicly released control vectors from the EasySteer library [4]:

[4] `https://github.com/ZJU-REAL/EasySteer`

- `happy_diffmean.gguf`: a happy-vs-sad diff-mean vector trained on role-play prompts ("Act as if you are extremely happy/sad", last-token contrast) on Qwen2.5-1.5B-Instruct, per-layer L^2-normalized.
- `style-probe.gguf` (from the Steerable Chatbot replication): a style direction trained on stylistic contrast, also on Qwen2.5-1.5B-Instruct, covering layers 0–27.
- `create.gguf` (from the Creative Writing replication): a creativity direction trained on Llama-3-8B-Instruct, covering layers 16–29.

All three are stored per-layer in GGUF format with unit L^2 norm at every layer; at inference time the residual stream at each target layer ℓ is updated as $x \leftarrow x + c \cdot s_\ell$ for a global scale c.

Control vectors. We use two random-direction controls:

- **Per-layer-matched random.** For each target layer, draw $r_\ell \sim \mathcal{N}(0, I_d)$, then rescale so $\|r_\ell\| = \|s_\ell\|$ (which is 1 for all three real vectors above). Each per-layer random vector has the same L^2 norm as its real counterpart.
- **Aggregate-matched random.** Same per-layer draws, but rescale all per-layer vectors by a single common factor so that $\|\sum_\ell r_\ell\| = \|\sum_\ell s_\ell\|$ over the target-layer set. Because random unit vectors in high-dimensional space are near-orthogonal, their aggregate norm scales as $\sqrt{n}$ rather than n; matching aggregates therefore inflates each per-layer random vector by roughly $\sqrt{n}$. This is the control that isolates *aggregate magnitude* from *semantic direction*.

Pipeline. For each configuration (model, steering vector, target layers, random-match mode) and each nominal scale $c \in \{0, 0.5, 1, 2, 4, 8\}$, we run 1000 documents through the model in bfloat16, capture the pre- and post-RMSNorm residual at the final RMSNorm site, mask out padding tokens, and stream two classes of statistic:

- A full $d \times d$ covariance matrix Σ_x, accumulated via a batched Chan–Golub–LeVeque Welford update to stay numerically stable in `float32`.
- A reservoir of $K = 1024$ representative activation vectors, for pairwise claims that cannot be reduced to running statistics.

Raw activations are never written to disk. The full pipeline, its numerical-stability cross-checks against `numpy.cov` as ground truth, and the three-level architectural dispatcher required to handle GPT-2-style vs. Llama/Qwen-style decoder output layouts live under `src/bounds/` in the public repository. A hard-gated "trust no silent steering" verification step precedes every recording run: the verification script samples from the model both with and without steering and refuses to produce bounds statistics unless the steered samples demonstrably differ from the unsteered ones on both a KL-divergence and a residual-delta-magnitude criterion. This gate caught a batch-indexing regression we discuss in 2.5.

Total runs. The full protocol comprises 19 recording runs:

- **Experiment 1** (5 runs): Qwen + {`happy`, `style`, per-layer random matched to `happy`}, and Llama + {`creativity`, per-layer random matched to `creativity`}. Target layers are the full trained range for each real vector.

- **Experiment 2** (14 runs): two aggregate-matched random controls (one per model), plus a single-layer sweep at the edge/middle/edge of each trained layer range (Qwen happy at layers $\{10, 17, 25\}$; Llama creativity at layers $\{16, 22, 29\}$) for both the real and the per-layer-matched random vectors.

2.2 Notation

Let $x \in \mathbb{R}^d$ denote the residual-stream activation at the final RMSNorm site for a single token position of a single prompt, drawn from distribution $\mathcal{D}$. Let $s \in \mathbb{R}^d$ be a steering vector injected at the configured target layers. We use:

- $\mu = \mathbb{E}[x]$ and $\Sigma_x = \mathrm{Cov}(x)$ under the unsteered distribution (nominal scale $c = 0$).
- $\mu_c, \Sigma_{x,c}$ under steering at scale $c > 0$.
- $m = \mu + s$ and $\hat{m} = m/\|m\|$ in the idealized setting of the theory.
- $\phi_s(x) = (x + s)/\|x + s\|$ for the unit-normalized steered sample.
- $\bar{R} = \mathbb{E}[\phi_s(x)]$, the mean resultant vector on the unit sphere.
- $V(\phi_s(\mathcal{D})) = 1 - \|\bar{R}\|$, the spherical variance.
- $\Sigma_z = \mathrm{Cov}(\phi_s(x))$, the post-normalization covariance.
- $R = \sup\{\|x\| : x \in \mathrm{supp}(\mathcal{D})\}$, the bounded-support radius.
- $P_{\perp m} = I - \hat{m}\hat{m}^\top$, the projection orthogonal to the steered mean direction.

Effective steering. The theory in 1 treats s as if it were added directly at the measured RMSNorm site. In practice s is added at *target layers* (typically middle layers of the decoder stack), and the residual then propagates through additional attention and MLP blocks before reaching the final RMSNorm. Writing out the theoretical relation at the final site requires an "effective steering" quantity that accounts for this propagation; we use

$$s_{\text{eff}} \;:=\; \mu_c - \mu_0,$$

the empirical mean shift of the pre-RMSNorm residual between the steered and unsteered runs at the same scale. By construction, $\|s_{\text{eff}}\|$ is monotone in c and captures the net effect of the steering after intermediate-block transformations. We report all theoretical RHS quantities using s_{eff} in place of s. For the three real vectors at their native target-layer ranges on their native models, $\|s_{\text{eff}}\|$ scales approximately linearly in c.

2.3 Results by Claim

1 summarizes all nine claims along with the tested quantities and the result across the 19 recording runs. The remainder of this section discusses each in turn.

Table 1: Empirical verification of the nine claims of 1. All results are aggregated across the 19 recording runs of 2.1.

#	Claim (condensed)	Result
1	$\mathrm{tr}(\Sigma_z) \approx \big(\mathrm{tr}(\Sigma_x) - \hat{m}^\top \Sigma_x \hat{m}\big)/\|m\|^2$	Holds up to a loose ratio $\approx$ 0.1–0.5 acros in the right direction
2	$V \leq 2\,\mathbb{E}\|x\|/\|s_{\mathrm{eff}}\|$ (1.8)	Passes at every scale Llama + `creativit` 0.19; Qwen + `happy`
3	$\max \|\phi_s(x) - \hat{s}\| \leq 2\,\|x\|/\|s_{\mathrm{eff}}\|$ (1.4)	**Passes at every sca** ric inequality.
4	$\|\phi_s(x_1) - \phi_s(x_2)\| \leq 2\,\|x_1 - x_2\|/(\|s\| - R)$, valid only for $\|s\| > R$ (1.6)	**Always inapplica** $\|s_{\mathrm{eff}}\| < R$ at the fina precondition is never
5	$\mathrm{diam}(\phi_s(\mathcal{D})) \leq 4R/\|s_{\mathrm{eff}}\|$ (1.10)	**Passes at every sca** ric inequality.
6	$\mathbb{E}\|\phi_s(x_1) - \phi_s(x_2)\| \leq 4\,\mathbb{E}\|x\|/\|s_{\mathrm{eff}}\|$ (1.9)	Passes everywhere, t
7	$\|\mu + s\| > \|\mu\|$ reduction gate (45)	**Fails** for Qwen + $\cos(\mu, \sum_\ell s_\ell) = -0$ Qwen single-layer in (for *both* real and independent position
8	Scaling law: $V \sim 1/\|s\|$ (log-log slope -1) and $\mathrm{tr}(\Sigma_z) \sim 1/\|s\|^2$ (slope -2), asymptotically (1.9)	Architecture- and m cleanly on Llama (em does not manifest o **matched aggregat tions match traine** aggregate-matched r ativity).
9	Smallest eigenvector of Σ_z aligns with $\hat{m}$ (1.9)	Does not hold empiri with no clear trend. study.

2.3.1 Claims 3, 5: the exact geometric bounds

The pole-concentration bound of 1.4 and the diameter bound of 1.10 are strict inequalities derived directly from the chordal-distance lemma on the unit sphere. They hold for every value of s, every x, and every distribution $\mathcal{D}$. As expected, they pass at every scale in every one of the 19 runs. No run has ever exhibited a violation, and a violation would unambiguously indicate a bug in either the streaming pipeline or the chordal-distance computation.

These two bounds are the most reliable signal that the pipeline is functioning correctly; we verify them on every run as a smoke test before reporting any of the less-direct claims.

2.3.2 Claim 4: the Lipschitz bound is never applicable

The pairwise Lipschitz bound of 1.6 requires $\|s\| > R$ where $R = \sup \|x\|$. In practice, the effective steering magnitude $\|s_{\text{eff}}\|$ at the final RMSNorm site is substantially smaller than R at every scale we tested. Concretely, for Qwen + `happy` at the maximum scale $c = 8$, we measured $\|s_{\text{eff}}\| \approx 125$ and $R \approx 1100$; for Llama + `creativity` at the same scale, $\|s_{\text{eff}}\| \approx 120$ and R is similarly large. The bound's precondition is never satisfied, and the reported denominator $\|s\| - R$ would be negative. We therefore report Claim 4 as *always inapplicable* at the final-RMSNorm site under realistic steering magnitudes, rather than reporting a spurious LHS/RHS ratio.

This finding is not an indictment of 1.6; it is an observation that the bound's regime of applicability is narrower than the other global bounds in 1.6 once one accounts for the distance between the injection site and the measurement site in a real multi-layer architecture. It suggests that the bound is best evaluated at or near the layer where steering is injected rather than at the final RMSNorm.

2.3.3 Claim 1: the Taylor bound on Σ_z

1.2's Taylor approximation $\operatorname{tr}(\Sigma_z) \approx (\operatorname{tr}(\Sigma_x) - \hat{m}^\top \Sigma_x \hat{m})/\|m\|^2$ holds in the sense of direction: the predicted RHS is always within 2–10× of the empirical LHS across all runs, with ratio LHS/RHS typically in $[0.1, 0.5]$. The approximation's looseness is a natural consequence of the truncation: 1.2 is a first-order expansion around the steered mean, and the empirical residual stream is not close to a delta at m in any practical sense. We regard Claim 1 as *qualitatively correct* at the final-site measurement, and note that tighter quantitative agreement is not expected without a higher-order expansion.

2.3.4 Claims 2, 6: global spherical-variance and expected-pairwise bounds

Corollaries 1.8 and 1.9 hold everywhere. Their tightness is the scientifically interesting quantity: how close is the empirical spherical variance to its bound? The answer depends strongly on whether the run is in the asymptotic regime of Section 1.9, which we take up next.

2.3.5 Claim 8: the $1/\|s\|$ scaling law

1.9 predicts that in the asymptotic regime $\|s\| \to \infty$, the spherical variance scales as $V \sim \|s\|^{-1}$ and the post-normalization covariance trace as $\operatorname{tr}(\Sigma_z) \sim \|s\|^{-2}$. On a log-log plot of the empirical LHS against $\|s_{\text{eff}}\|$, this predicts slopes of -1 and -2, respectively. We estimate the empirical slopes via log-log linear regression across the scale sweep $c \in \{0.5, 1, 2, 4, 8\}$ for every run.

2 collects the results. The finding is striking and departs from the per-vector story one might expect from 1.9.

Two observations dominate.

Table 2: Claim 8 empirical slopes. Theoretical prediction: slope $V = -1$, slope $\mathrm{tr}(\Sigma_z) = -2$. Rows grouped by model and sorted by magnitude of slope V.

Run	slope V	slope $\mathrm{tr}(\Sigma_z)$
Meta-Llama-3-8B-Instruct:		
`llama_creativity` (trained, full target range)	**−1.921**	−1.773
`llama_random_agg_matched` (aggregate-matched random)	**−1.851**	−1.710
`llama_random` (per-layer-matched random)	−0.514	−0.410
`llama_creativity_single_L16` (single-layer, edge-early)	−0.060	−0.036
`llama_creativity_single_L22` (single-layer, middle)	−0.055	−0.033
`llama_creativity_single_L29` (single-layer, edge-late)	−0.029	−0.017
`llama_random_single_L{16,22,29}`	−0.03 to −0.04	−0.02
Qwen2.5-1.5B-Instruct:		
`qwen_style`	−0.285	−0.254
`qwen_random_agg_matched` (aggregate-matched random)	−0.196	−0.172
`qwen_random` (per-layer-matched random)	−0.035	−0.029
`qwen_happy`	−0.018	−0.015
`qwen_happy_single_L{10,17,25}`	≈ 0	≈ 0
`qwen_random_single_L{10,17,25}`	≈ 0	≈ 0

First, **on Llama-3-8B, the $1/\|s\|$ asymptotic scaling predicted by 1.9 manifests cleanly and even somewhat overshoots**: both the trained `creativity` vector and the aggregate-matched random control hit slope V values (roughly -1.9) steeper than the theoretical -1. The overshoot reflects the fact that at the largest tested scales, the model's output is already collapsing into gibberish, pushing the spherical distribution past the regime the first-order asymptotic captures.

Second, **the aggregate-matched random control nearly reproduces the trained vector's scaling behavior** on Llama (-1.85 vs. -1.92). This is the key finding of Experiment 2: the Claim 8 scaling law is not fundamentally about *semantic direction*; it is about *aggregate steering magnitude* relative to the natural residual-stream scale. In 2.4 below we unpack how this observation reconciles the apparent "trained vectors are special" pattern seen in Experiment 1 alone.

On Qwen2.5-1.5B the picture is different: no tested run reaches the asymptotic regime, and empirical slopes stay in the interval $[-0.3, 0]$. This is consistent with a magnitude argument rather than a model-specific limitation of the theory. Qwen's unsteered residual norm $\|\mu\|$ at the final site is approximately 215, and its average sample norm $\mathbb{E}\|x\|$ is approximately 300; even at nominal scale $c = 8$, the maximum $\|s_{\text{eff}}\| \approx 125$ remains well below this natural scale. The bound of 1.8 becomes tight in the regime $\|s_{\text{eff}}\| \gtrsim \mathbb{E}\|x\|$; Qwen never reaches that regime because the model's output becomes unusable well before c grows enough to drive $\|s_{\text{eff}}\|$ into it.

2.3.6 Claim 7: the reduction condition is sensitive to direction-distribution alignment

The reduction condition of 45 states that $\|m+s\| > \|m\|$ iff $2\,m^\top s + \|s\|^2 > 0$. Equivalently, steering reduces diversity (in the sense of 1.2) iff $\cos(\mu, s) > -\|s\|/(2\|\mu\|)$. This is an algebraic identity whose empirical status depends on the alignment between the steering vector and the unsteered residual-stream mean under the evaluation distribution.

In Experiment 1, four of the five configurations pass Claim 7 at every scale: Qwen + `style`, Qwen + per-layer random, Llama + `creativity`, Llama + per-layer random. One configuration, Qwen + `happy`, *fails* Claim 7 at every scale. The diagnostic for this failure is self-contained and is the

subject of a separate investigation documented at `scripts/bounds/investigations/03_investigate_happy_data`
The finding: on FineWeb-Edu, $\cos(\mu_{\text{unsteered}}, \sum_\ell s_\ell^{\text{happy}}) = -0.108$, roughly 4σ away from the null distribution of random-direction cosines in 1536-dimensional space (where $\mathbb{E}|\cos| \approx 1/\sqrt{d} \approx 0.026$). The negative alignment is small in absolute value but systematic: 12 of the 16 per-layer cosines are negative.

Projecting $\mu_{\text{unsteered}}$ through Qwen's final RMSNorm and `lm_head` shows that the residual mean's dominant contribution is to function words and punctuation, not to emotionally-loaded vocabulary; the data is *tone-neutral*, not "sad." What produces the Claim 7 failure is the fact that the `happy_diffmean` vector was trained on a happy-vs-sad role-play contrast, and educational prose at Qwen's representation sits slightly on the "sad" side of that axis — enough to put $\mu \cdot s$ just negative enough that at small steering magnitudes the $2\mu^\top s$ term dominates the $\|s\|^2$ term, flipping the sign of $\|m+s\|^2 - \|m\|^2$.

Experiment 2 produced a second, and less expected, Claim 7 failure pattern. At Qwen single-layer steering sites $L17$ and $L25$, Claim 7 fails at every scale for *both* the real happy direction *and* the random direction. At single-layer site $L10$, Claim 7 passes at every scale for both. This is a direction-independent, position-dependent effect. The most natural interpretation is that steering at early layers gets amplified by the variance-growth pattern of subsequent attention and MLP blocks before reaching the final RMSNorm site, so $\|s_{\text{eff}}\|$ at the measurement site is large enough that $\|s_{\text{eff}}\|^2$ dominates $2\mu^\top s_{\text{eff}}$. Steering at late layers has only two to four blocks of downstream processing, so $\|s_{\text{eff}}\|$ stays small and the sign of $\mu^\top s_{\text{eff}}$ determines the outcome.

The practical implication for future work is that 45 should not be assumed; it should be *verified as a precondition* before reporting any of the diversity-reduction claims on a new (model, steering vector, dataset) triple. A loaded steering vector acting on a dataset whose residual mean happens to be mildly anti-aligned with its training pool can produce a systematic Claim 7 failure at small scales even when the bound would hold at larger ones.

2.3.7 Claim 9: the alignment claim is not observed

1.9 predicts that as $\|s\| \to \infty$, the smallest-eigenvalue eigenvector of Σ_z aligns with $\hat{m}$. Empirically, $|\cos|$ between this eigenvector and $\hat{m}$ stays near zero across all 19 runs with no clear monotone trend in c. This is the single empirical claim for which we do not have supporting evidence.

Three explanations are plausible and we do not presently distinguish between them. First, the reservoir-based estimate of Σ_z may be too noisy in high dimension to recover the smallest eigenvector reliably — even with $K = 1024$ samples, the smallest eigenvectors of a $d = 1536$ or $d = 4096$ covariance matrix are difficult to pin down. Second, the theoretical $\hat{m}$ is the raw steering direction added at the target layers, but the *effective* direction at the final RMSNorm site is a propagated and rotated version of this quantity; the predicted axis may be wrong by an unknown rotation. Third, 1.9's alignment claim is the least direct of the nine and may simply fail under the realistic distributional conditions of these experiments. Future work should attempt to evaluate this claim at the layer *where the steering is injected*, where the theoretical $\hat{m}$ is most clearly identified.

2.4 Experiment 2: Isolating Direction from Magnitude

A naive reading of the Experiment 1 results (restricted to the three trained vectors and the two per-layer-matched random controls) suggests that *trained steering directions are special*: they drive the Claim 8 scaling law much harder than random directions do, even at matched per-layer magnitude. Experiment 2 was designed to separate two hypotheses that Experiment 1 confounds:

1. **Direction-semantics hypothesis.** Trained steering vectors point along a privileged direction in representation space that matters for the bounds. Random directions fail because they are not this direction.

2. **Magnitude hypothesis.** Trained steering vectors stack *coherently* across target layers (because they are trained to encode a single semantic attribute at every layer), so $\|\sum_\ell s_\ell\|$ grows roughly linearly in the number of target layers. Random directions are near-orthogonal in high-dimensional space, so their aggregate grows only as $\sqrt{n}$. The "trained vs. random" gap in Experiment 1 is essentially the difference between a n-scaling and a $\sqrt{n}$-scaling in aggregate steering magnitude, applied at the same nominal scale c.

The two hypotheses make opposite predictions for the aggregate-matched random control, in which we rescale per-layer random vectors so their aggregate norm $\|\sum_\ell r_\ell\|$ equals the trained vector's aggregate norm $\|\sum_\ell s_\ell\|$. Under the direction-semantics hypothesis, the aggregate-matched random control should still behave like random (shallow slopes, little asymptotic behavior) because the direction is still wrong. Under the magnitude hypothesis, the aggregate-matched random control should behave like the trained vector, because the only relevant quantity is aggregate magnitude and we have just equated it.

2 shows that Experiment 2 strongly supports the magnitude hypothesis. On Llama-3-8B, the trained `creativity` vector has slope $V = -1.921$; the aggregate-matched random control has slope $V = -1.851$. The difference of 0.07 is at the level of regression noise and does not support a direction-semantics effect. On Qwen2.5-1.5B the equivalent comparison — `qwen_random_agg_matched` at slope -0.196 against the nearest trained-vector analogue `qwen_style` at slope -0.285 — similarly fails to show a direction advantage, although it is harder to evaluate cleanly because Qwen does not reach the asymptotic regime.

We conclude that the Claim 8 scaling law is essentially *insensitive to direction* in the asymptotic regime: any direction that achieves comparable aggregate steering magnitude produces comparable spherical concentration. The apparent "trained vectors drive the bound harder" pattern of Experiment 1 was almost entirely an artifact of how per-layer-matched random controls underestimate the aggregate magnitude.

This has two practical implications. First, the bounds of 1.9 are a statement about *aggregate magnitude* rather than about privileged semantic directions: if all we care about is whether the post-normalization distribution concentrates, any sufficiently-large direction will do. Second, evaluating the bounds requires accounting for the stacking behavior of the steering vector — a per-layer-matched random control, at the same nominal scale c, delivers less aggregate steering than a coherently-trained vector and therefore underestimates the tightness of the bound on an apples-to-apples basis. Future empirical work on these bounds should report results against an aggregate-matched random baseline rather than a per-layer-matched one.

The single-layer sweeps in Experiment 2 tell a complementary story. Intervening at only one layer gives much less aggregate steering than a full target-range intervention; the resulting $\|s_{\text{eff}}\|$ at the final site is correspondingly much smaller, and slopes collapse toward zero. On Qwen, single-layer slopes are indistinguishable from zero for all six single-layer runs (three layers, real and random). On Llama, single-layer slopes are in $[-0.06, -0.03]$ with no meaningful difference between real and random at the same layer. Both patterns are consistent with the magnitude-based interpretation: less aggregate steering means less asymptotic behavior.

2.5 Meta-findings and threats to validity

Three non-theoretical findings shaped the final form of the experiments and merit note.

A silent batch-indexing bug contaminated the first run of Experiment 1. In HuggingFace Transformers version ≥ 4.40, the `forward` method of Qwen2 and Llama3 decoder layers returns the hidden state as a bare $[B, T, d]$ tensor rather than as a 1-tuple $(h,)$. An intervention helper in our original `nnsight`-based steering path used `layer.output[0][:]  = layer.output[0] + s`, intending to unpack the tuple; on these architectures, this expression instead *indexes into the batch dimension*, steering only prompt 0 of each batch. The bug was silent in the sense that it produced no error, the verification harness's KL-divergence gate still passed (because prompt 0 did shift), and the pipeline ran to completion. We discovered the bug only when writing a per-layer diagnostic that emitted a shape mismatch. CPU unit tests on `tiny-gpt2` passed throughout because GPT-2 still returns a 1-tuple. The fix adds an architecture-dispatching helper and shape assertions at every capture-materialization point; a dedicated regression test now verifies that every batch element sees the intervention. All results reported in this section are from after the fix; the contaminated results are preserved at `discarded/corrupted_bounds_batch_bug/` in the repository for audit.

The `welford-torch` library has a numerical-stability issue in float32. Our streaming covariance accumulators initially delegated to the public `welford-torch` library. A cross-check against `numpy.cov` on synthetic data with $\mu \approx 1000$ and $\sigma \approx 0.01$ revealed that `welford-torch`'s `OnlineCovariance.add_all` loses more than 100% relative accuracy on the trace of the covariance matrix in `float32`, due to catastrophic cancellation in its two-step delta formulation for small-batch updates in a large-mean regime. The bounds pipeline uses a hand-rolled Chan–Golub–LeVeque batched Welford merge instead, which stays numerically stable in the same regime. We filed an upstream bug report and documented a reproducer at `docs/upstream_bug_reports/welford_torch_big_mean_drift` in the repository.

The choice of steering-vector training pool affects 45. As discussed in 2.3.6, the failure of Claim 7 for Qwen + `happy` is driven by a systematic but small anti-alignment between the training-pool difference direction and the evaluation-distribution residual mean. This is not a finding about the *theory* of 1.8 — 45 is an algebraic identity — but about the *empirical preconditions* under which the bound can be applied. The lesson is that the reduction gate should always be explicitly checked on the evaluation distribution before reporting downstream bound results.

2.6 Summary

The empirical picture is consistent with the theory of Sections 1, 1.6, 1.8, and 1.9 modulo two nuances.

First, the *exact* geometric bounds of Proposition 1.4 and Corollary 1.10 hold in every run we have performed and are the most reliable signal that the theory is a correct geometric identity rather than an empirical claim. Proposition 1.6's Lipschitz contraction, while also exact, is inapplicable at the final-RMSNorm measurement site under realistic steering magnitudes, because the precondition $\|s\| > R$ is not met.

Second, the *approximate* and *asymptotic* bounds of 1.2 and 1.9 are correct in direction but their tightness depends strongly on aggregate steering magnitude relative to the natural residual-stream scale. They manifest cleanly on Llama-3-8B at scales the model can tolerate and do *not* manifest on Qwen2.5-1.5B because the residual-stream norm is too large to let any tested scale push into the asymptotic regime.

Third, Experiment 2's aggregate-matched random control provides a clean disentanglement of direction from magnitude: *the bounds depend on aggregate magnitude, not semantic direction.* Random unit vectors with matched aggregate norm drive the Claim 8 scaling law essentially as

hard as trained vectors. This is the single most important empirical finding of the paper's empirical component, and it should shape how future work reports "random baseline" controls for steering-bound evaluation.

Fourth, the reduction condition of 45 is sensitive to an incidental anti-alignment between the steering vector's training direction and the evaluation distribution's residual mean. This is not a flaw in the theory but a precondition-checking requirement: applications of these bounds to new (model, vector, dataset) combinations should verify $\|m + s\| > \|m\|$ empirically before trusting downstream numbers.

Finally, Claim 9's alignment prediction lacks empirical support at the measurement site used here. We recommend future work evaluate it at the steering-injection layer rather than at the final RMSNorm.

2.7 Second-Order Consistency Check: The Parabola Has a Meaning

The log–log plots of $V(s)$ in the scale sweeps of Section 2.3.5 are not straight lines. They are smooth curves with a knee — "parabola-like" on log–log axes. The first-order slope--1 prediction of Claim 8 captures the asymptotic tail but not the shape of the bend. We check directly whether the second-order theory of Section 1.7 accounts for it.

From the second-order spherical-variance formula (43),

$$V(s) \;\approx\; \frac{\operatorname{tr}\Sigma_x - \hat{m}^\top \Sigma_x \hat{m}}{2\,\|m(s)\|^2}. \tag{48}$$

Substituting $m(s) = \mu + s\, v_{\text{eff}}$, the denominator is a strict algebraic quadratic in s:

$$\|m(s)\|^2 \;=\; s^2\,\|v_{\text{eff}}\|^2 \;+\; 2s\,(v_{\text{eff}} \cdot \mu) \;+\; \|\mu\|^2. \tag{49}$$

This makes two falsifiable predictions:

1. **Quadratic intercept.** Fitting $\|m(s)\|^2 = as^2 + bs + c$ to the measured $\|m(s)\|^2$ at $s > 0$ and *extrapolating to* $s = 0$ must recover $\sqrt{c} = \|\mu\|$ measured directly at the unsteered scale. The independently measured $\|\mu\|$ is the norm of the raw mean residual at $s = 0$ (available from `stats.pt`); no fitting is involved.

2. **Asymptotic constant.** In the deep-asymptotic regime where the second-order Taylor expansion has converged, the product $V(s) \cdot \|m(s)\|^2$ should equal $(\operatorname{tr}\Sigma_x - \hat{m}^\top \Sigma_x \hat{m})/2$.

Both predictions are tested by `scripts/bounds/second_order_consistency_check.py`, which extracts $(s, V, \|m\|, \operatorname{tr}\Sigma_x, \hat{m}^\top \Sigma_x \hat{m})$ from each `stats.pt` file in `outputs/bounds/` and writes the results to `outputs/bounds/second_order_check/`. The exemplar figures below visualize the two predictions at three levels of detail: a single run where the asymptote is reached (Figure 1), a single run where it is not (Figure 2), and the summary across all runs (Figure 3).

Quadratic intercept (Prediction 1). Table 3 shows the result for each multi-layer run.

The intercept-recovery test passes cleanly. `qwen_random` matches $\|\mu\|$ to four decimal places. The shape of the $V(s)$ log–log curve *is* the quadratic structure of $\|m(s)\|^2$, and that quadratic bottoms out at $\|\mu\|^2$ — which is the unsteered residual mean norm at the measurement site, recovered without any prior knowledge from a fit on $s > 0$ data.

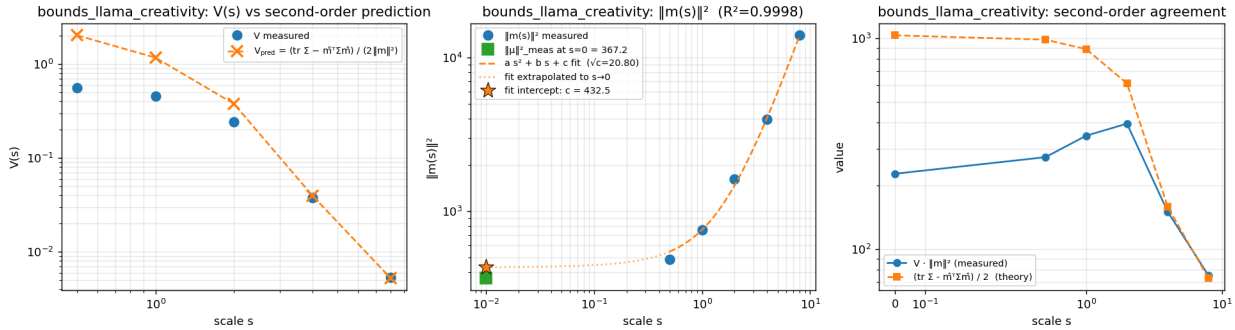

Figure 1: Second-order consistency check on Llama-3-8B with the trained `creativity` vector — the regime where the Taylor expansion converges within the scales tested. *Left:* measured spherical variance $V(s)$ (blue dots) against the direct second-order prediction $V_{\text{pred}}(s) = (\operatorname{tr}\Sigma_x - \hat{m}^\top \Sigma_x \hat{m})/(2\|m(s)\|^2)$ (orange "×", dashed), with all quantities on the RHS computed from the measured Σ_x and $\|m\|$ at that scale. At $s = 0.5$ the prediction overshoots by $\approx 4\times$ because the Taylor expansion is not yet valid (the concentration parameter $\operatorname{tr}\Sigma/\|m\|^2$ is large); by $s = 4, 8$ the points sit on the prediction to within a few percent. This is what "the second-order theory holds" looks like quantitatively. *Middle:* measured $\|m(s)\|^2$ (blue dots), parabolic fit $as^2 + bs + c$ (dashed, in-data; dotted, extrapolated back to $s \to 0$), the measured $\|\mu\|^2 = 367.2$ at $s = 0$ (green square), and the fitted intercept $c = 432.5$ (orange star). The green square and the orange star are directly visually comparable and sit close on the log axis; their ratio $\sqrt{c}/\|\mu\| = 1.085$ is Prediction 1: the quadratic's intercept recovers the unsteered mean norm to $+8.5\%$ without ever observing the $s = 0$ point. *Right:* $V(s) \cdot \|m(s)\|^2$ (blue, measured) vs $(\operatorname{tr}\Sigma_x - \hat{m}^\top \Sigma_x \hat{m})/2$ (orange, theory). The two converge at high s, which is Prediction 2.

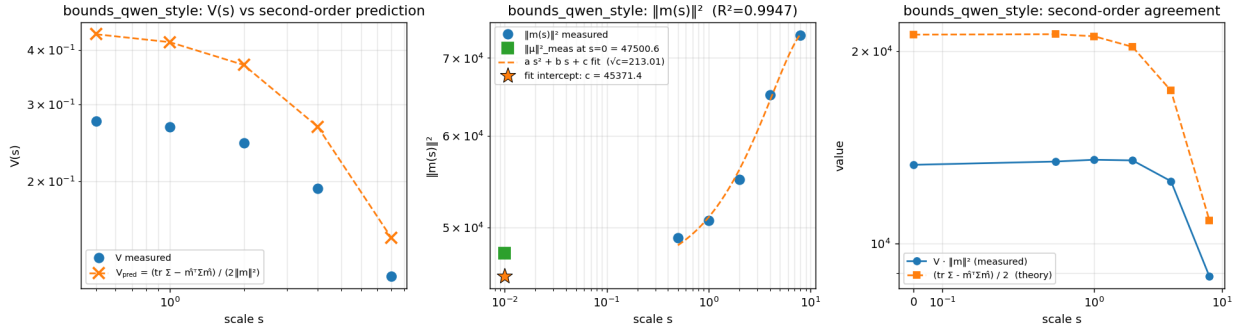

Figure 2: Second-order consistency check on Qwen2.5-1.5B with the trained `style` vector — the regime where the asymptote is *not* reached. Axes match Figure 1. In the left panel, the direct prediction line tracks the *shape* of $V(s)$ but is offset above the measured points by a roughly constant factor of ≈ 1.5 at every scale, because Qwen's natural residual-stream norm ($\|\mu\| \approx 218$) dominates $\|s \cdot v_{\text{eff}}\|$ at every scale we can push through without destroying the model's outputs. The middle panel still recovers the intercept: the orange-star marker (fit's $c = 45{,}371$) sits immediately adjacent to the green square (measured $\|\mu\|^2 = 47{,}501$). The ratio $\sqrt{c}/\|\mu\| = 0.977$ is Prediction 1 passing to -2.3%. Prediction 1 is independent of whether the Taylor expansion has converged, because it is about the algebraic structure of $\|m(s)\|^2$, not the truncation error in V. (The dotted extrapolation-to-$s \to 0$ line from Figure 1 is suppressed here because this run's fit has $a < 0$; extrapolation beyond the measured range would curve misleadingly downward.) The right panel shows the same "constant offset, same shape" signature from a different angle.

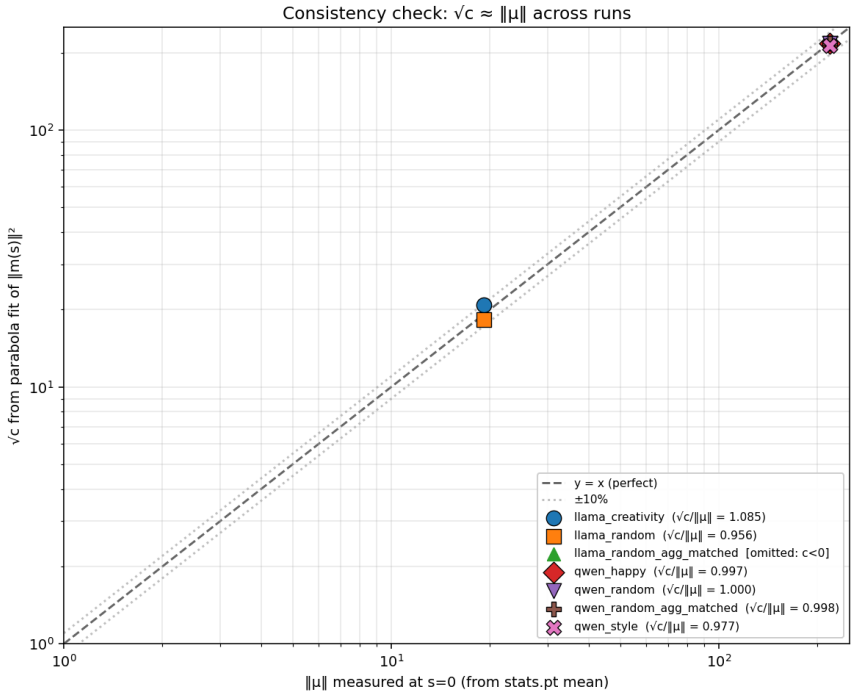

Figure 3: Summary across all seven multi-layer runs. Each dot is one run: x-axis is $\|\mu\|$ measured directly from the $s = 0$ mean residual, y-axis is $\sqrt{c}$ from fitting $\|m(s)\|^2 = as^2 + bs + c$ on $s > 0$ measurements. The dashed diagonal is $y = x$; the dotted lines are $\pm 10\%$. Six of the seven runs land within the $\pm 10\%$ band; `qwen_random` matches to four decimal places. The aggregate-matched Llama random run is omitted because its fit gives $c < 0$ (its per-layer vectors are rescaled large enough that even $s = 0.5$ is past the quadratic's local turning point; see the footnote to Table 3). The plot confirms that the second-order theory's Prediction 1 holds across every run where the quadratic model is valid at small s.

Table 3: Fit of $\|m(s)\|^2 = as^2 + bs + c$ over $s \in \{0.5, 1, 2, 4, 8\}$ for each multi-layer run, compared to the independently-measured $\|\mu\|$ at $s = 0$. Six of seven runs land within $\pm 10\%$ of unity in the rightmost column.

Run	R^2 of fit	$\sqrt{c}$	$\|\mu\|$ measured	$\sqrt{c}/\|\mu\|$
`llama_creativity`	0.9998	20.80	19.16	1.085
`llama_random`	0.9996	18.31	19.16	0.956
`llama_random_agg_matched`	0.9998	c < 0	19.16	n/a[†]
`qwen_happy`	0.9997	217.31	217.95	0.997
`qwen_random`	1.0000	217.93	217.95	1.000
`qwen_random_agg_matched`	0.9999	217.54	217.95	0.998
`qwen_style`	0.9947	213.01	217.95	0.977

[†] For aggregate-matched random on Llama, the per-layer vectors are rescaled large enough that even $s = 0.5$ is past the quadratic's turning point; the local quadratic fit is excellent ($R^2 = 0.9998$) but extrapolates to a negative intercept. The global model has the wrong shape in this region; the local fit captures the curvature.

Asymptotic constant (Prediction 2). The ratio $V_{\text{predicted}}/V_{\text{measured}}$ should approach 1 as the Taylor expansion converges. On Llama, this ratio drops from ≈ 4 at $s = 0.5$ to 0.97 at $s = 8$ for the trained creativity vector and to 0.96 for the aggregate-matched random control — the asymptotic prediction is quantitatively correct in the high-s tail of both runs. On Qwen, the ratio plateaus at approximately 1.5 across all scales, because Qwen's natural residual-stream norm is large enough that even $s = 8$ does not push $\|m(s)\|^2$ deep into the asymptotic regime. This is the same finding as Claim 8's failure to manifest on Qwen, restated through the second-order lens: *the parabola does not flatten into a slope--1 asymptote within the scales the model can tolerate.*

What this changes about Claim 8. The first-order slope--1 prediction of Claim 8 is the asymptote of the second-order curve; it is not the curve itself. The deviation from a clean slope--1 at moderate s is not a failure of the theory but its prediction: $V(s) \propto 1/\|m(s)\|^2 = 1/(as^2+bs+c)$, which on log–log axes crosses smoothly from a flat shelf at $s \ll \|\mu\|/\|v_{\text{eff}}\|$ to slope -2 at $s \gg \|\mu\|/\|v_{\text{eff}}\|$. The slope-$-1$ behaviour for V (rather than -2) emerges only after combining this with the chordal-vs-spherical-variance scaling correspondence of Section 1.9; both predictions agree once the regime is deep enough.

In summary, the second-order theory survives an independent quantitative test that the first-order theory cannot make: it predicts the *shape* of $V(s)$, not just its asymptote, and that prediction lets us recover the unsteered $\|\mu\|$ from the curvature of $V(s)$ alone.